



AWS Certificate Manager (ACM)

Lecture Slides

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Good luck on your exam!

Amazon Certificate Manager (ACM)



Provision, **manage**, and **deploy** SSL/TLS certificates for AWS services

Introduction to ACM

Amazon Certificate Manager simplifies the process of creating, managing, and deploying **SSL/TLS** certificates to ensure secure communications for your AWS-hosted websites and applications.

ACM handles 🙌 two kinds of certificates

1. **Public** – provided by ACM (Free)
2. **Private** – imported by user (\$400 / month)!!!

ACM can handle multiple subdomains
and wildcard domains (*)



The screenshot shows a table with two rows of domain names. The first row has 'exampro.co' and a 'Remove' button. The second row has '*.exampro.co' and a close button (X).

Domain name*	Remove
exampro.co	
*.exampro.co	X

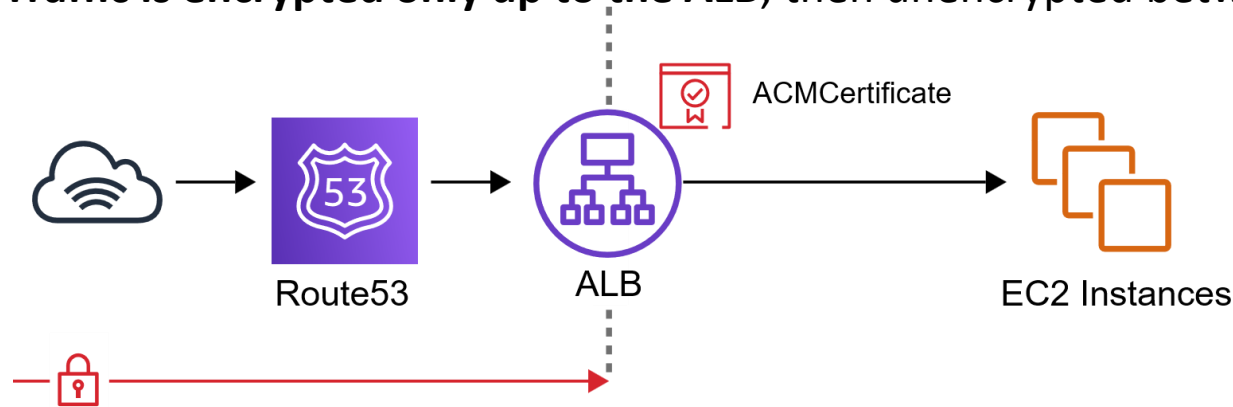
ACM can be **attached** to:

- Elastic Load Balancer (ELB)
- Amazon CloudFront
- API Gateway
- Elastic Beanstalk (via ELB)

ACM Example

1. SSL Termination at the Load Balancer

Traffic is encrypted only up to the ALB, then unencrypted between the ALB and EC2 instances.



Uses **ACM certificate** at the ALB only

Pros:

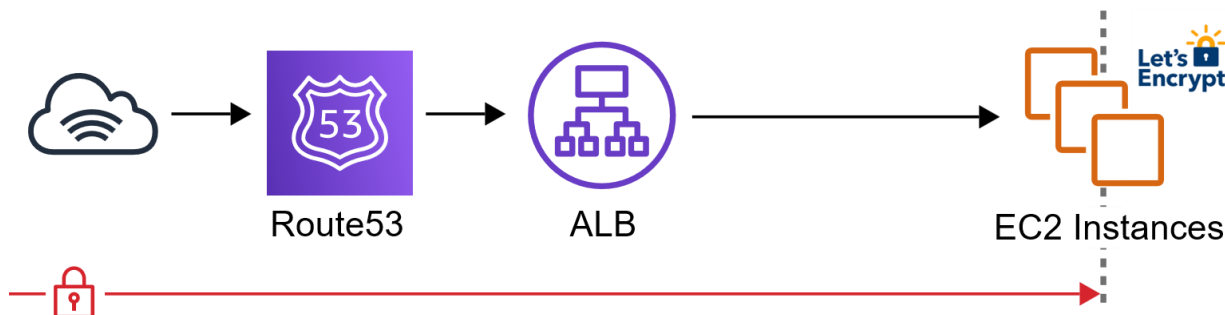
- Easy to manage
- No need to install certs on individual instances
- Scales easily with more EC2 instances

Cons:

- **Less secure** due to unencrypted traffic beyond the ALB

2. SSL End-to-End Encryption

Traffic remains encrypted from the client to the EC2 instance.



Uses an additional certificate (e.g., **Let's Encrypt** on each EC2 instance

Pros:

- **Ensures full end-to-end encryption**

Cons:

- More complex certificate management
- Requires cert installation and renewal on each EC2 instance